Triformer: Triangular, Variable-Specific Attentions for Long Sequence Multivariate Time Series Forecasting

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Abstract

A variety of real-world applications rely on far future information to make decisions, thus calling for *efficient* and *accurate* long sequence multivariate time series forecasting. While recent attention-based forecasting models show strong abilities in capturing long-term dependencies, they still suffer from two key limitations. First, canonical self attention has a quadratic complexity w.r.t. the input time series length, thus falling short in *efficiency*. Second, different variables' time series often have distinct temporal dynamics, which existing studies fail to capture, as they use the same model parameter space, e.g., projection matrices, for all variables' time series, thus falling short in *accuracy*.

To ensure high *efficiency* and *accuracy*, we propose `Triformer`, a triangular, variable-specific attention. (i) *Linear complexity*: we introduce a novel patch attention with linear complexity. When stacking multiple layers of the patch attentions, a triangular structure is proposed such that the layer sizes shrink exponentially, thus maintaining linear complexity. (ii) *Variable-specific parameters*: we propose a light-weight method to enable distinct sets of model parameters for different variables' time series to enhance accuracy without compromising efficiency and memory usage. Strong empirical evidence on four datasets from multiple domains justifies our design choices, and it demonstrates that `Triformer` outperforms state-of-the-art methods w.r.t. both accuracy and efficiency. Source code is publicly available at https://github.com/razvanc92/triformer.

1 Introduction

Long sequence multivariate time series forecasting plays an essential role in planning and managing complex systems across diverse domains [Liu *et al.*, 2018; Guo *et al.*, 2020; Pedersen *et al.*, 2020a; Pedersen *et al.*, 2020b; Yang *et al.*, 2020]. In such settings, multiple sensors are often deployed to collect diverse information related to a complex system, thus giving rise to multivariate time series [Cirstea *et al.*, 2019]. Figure 1 shows an example of a 3-variate time series indicating the power consumption of three clients from a power grid system, where each variable has its own time series.

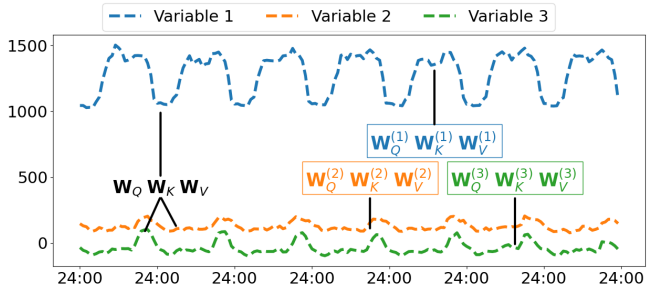

Figure 1: Example of 3-variate time series, where each variable has a distinct temporal pattern. **Variable-agnostic modeling:** same projection matrices (see the black matrices) are employed for all variables. **Variable-specific modeling:** distinct projection matrices (see the colorful matrices) are used for different variables, thus enabling the capture of distinct temporal patterns of different variables.

While extensive studies on short term forecasting exist, e.g., dozens steps ahead forecasting [Hochreiter and Schmidhuber, 1997; Qin *et al.*, 2017; Wu *et al.*, 2019; Wu *et al.*, 2020], a limited number of studies focus on long term forecasting, e.g., hundreds steps ahead. Recent studies show that attentions [Vaswani *et al.*, 2017; Wu *et al.*, 2021b] are able to capture better long term dependencies, comparing to Recurrent Neural Networks `RNN`s [Li *et al.*, 2018; Hu *et al.*, 2020; Kieu *et al.*, 2022a] or Temporal Convolutional Networks `TCN`s [Wu *et al.*, 2019; Campos *et al.*, 2022; Kieu *et al.*, 2022b]. However, two main limitations still exist.

High complexity: For a time series of H timestamps, canonical self-attention has a quadratic complexity of $\mathcal{O}(H^2)$. Recent studies on long-term forecasting propose different "sparse" versions of self-attentions [Zhou *et al.*, 2021; Kitaev *et al.*, 2020], aiming at reducing the high complexity. We propose `Triformer` with linear complexity $\mathcal{O}(H)$. We first break the time series into small patches. For each patch,

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we define a novel *Patch Attention* (`PA`) with linear complexity (cf. the small white triangles in Figure 2). Specifically, we introduce a pseudo timestamp for a patch and we compute attentions of the timestamps in the patch only to the single pseudo timestamp, making patch attention linear. Then, only the pseudo timestamps are fed into the next layer, such that the layer sizes shrink exponentially, making a triangular structure as shown in Figure 2. This ensures a multi-layer `Triformer` still have linear complexity $\mathcal{O}(H)$.

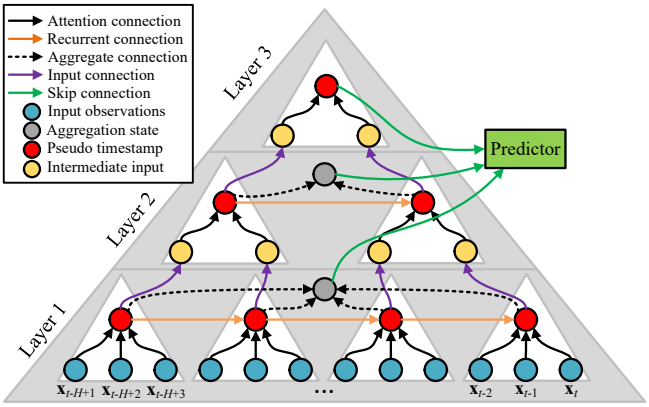


Figure 2: `Triformer` has a triangular structure as the layer input sizes shrink exponentially, making a multi-layer `Triformer` still with $\mathcal{O}(H)$.

Variable-agnostic parameters: Existing forecasting models often use variable-agnostic parameters, although different variables may exhibit distinct temporal patterns [Wu *et al.*, 2019; Li *et al.*, 2018; Wu *et al.*, 2021b]. For example, as shown in Figure 1, although variable 1 has a unique temporal pattern, which is different from the patterns of variables 2 and 3, the same model parameters, e.g., the projection matrices $\mathbf{W}_Q, \mathbf{W}_K$, and $\mathbf{W}_V$ used in self-attentions (cf. the black matrices in Figure 1), are applied to all three variables. This forces the learned parameters to capture an "average" pattern among all the variables, thus failing to capture distinct temporal pattern of each variable and hurting accuracy.

We propose a light-weight method to enable variable specific parameters, e.g., a distinct set of matrices $\mathbf{W}_Q^{(i)}, \mathbf{W}_K^{(i)}$, and $\mathbf{W}_V^{(i)}$ for the i-th variable (cf. the colorful matrices in Figure 1), such that it is possible to capture distinct temporal patterns for different variables. Specifically, we factorize the projection matrices into variable-agnostic and variable-specific matrices, where the former is shared among all variables and the latter is specific to different variables. We make the variable-specific matrices very compact, thus avoiding increasing the parameter space and computation overhead.

We summarize the main contributions as follows: (i) We propose a novel, efficient attention mechanism, namely Patch Attention, along with its triangular, multi-layer structure. This ensures an overall linear complexity, thus achieving high **efficiency**. (ii) We propose a light-weight approach to enable variable-specific model parameters, making it possible to capture distinct temporal patterns from different variables, thus enhancing **accuracy**. (iii) We conduct extensive experiments on four public, commonly used multivariate time series data sets from different domains, justifying our design choices and demonstrating that the proposal outperforms state-of-the-art methods.

2 Related Work

We categorize multivariate time series forecasting methods in Table 1 along two dimensions—short vs. long term forecasting and variable-agnostic vs. variable-specific modeling.

	Short Term	Long Term
Variable Agnostic	[Wu *et al.*, 2021a] [Cao *et al.*, 2020] [Bai *et al.*, 2019] [Wu *et al.*, 2019] [Li *et al.*, 2018]	[Zhou *et al.*, 2021] [Kitaev *et al.*, 2020] [Li *et al.*, 2019] [Stoller *et al.*, 2020] [Vaswani *et al.*, 2017]
Variable Specific	[Cirstea *et al.*, 2022b] [Cirstea *et al.*, 2021] [Bai *et al.*, 2020] [Pan *et al.*, 2019]	`Triformer`

Table 1: Categorization of Time Series Forecasting.

Short vs. Long Term Forecasting: Most studies focus on short-term forecasting, e.g., 12 to 48 steps ahead. Short-term forecasting models often rely on two types of models that are good at capturing short-term temporal dependencies—recurrent neural networks (`RNN`s), e.g., `LSTM` or `GRU` [Bai *et al.*, 2019; Li *et al.*, 2018; Seo *et al.*, 2018], and temporal convolutions networks (`TCN`s), e.g., 1D convolution and `WaveNet` [Cao *et al.*, 2020; Wu *et al.*, 2019]. Both `RNN`s and `TCN`s have limited capabilities when dealing with long-range dependencies since they can not directly access the whole input time series, as they rely on intermediate representations [Khandelwal *et al.*, 2018]. Thus, they fall short on long-term forecasting.

For long term forecasting, self-attention based models achieve superior accuracy, but they suffer from quadratic memory and runtime overhead w.r.t. the input time series length H. To reduce the complexity, recent long term forecasting studies propose sparse attentions: `LogTrans` [Li *et al.*, 2019] is with $\mathcal{O}(H(\log H)^2)$, and `Informer` [Zhou *et al.*, 2021] further reduces the complexity to $\mathcal{O}(H \log H)$. There also exist other efficient versions of attentions [Beltagy *et al.*, 2020; Wang *et al.*, 2020; Katharopoulos *et al.*, 2020], but they are not designed and verified for time series forecasting. When stacking multiple layers of attentions, an additional pooling layer is often used after each attention layer. This helps shrink the input size to the next attention layer, thus reducing overall complexity [Dai *et al.*, 2020; Zhou *et al.*, 2021]. We propose a novel efficient patch attention with linear complexity, which is able to shrink the layer size by itself without using additional pooling, making a multi-layer patch attention structure also linear.

Variable-agnostic vs. variable-specific modeling: Most studies are variable agnostic, meaning that, although different variables' time series may exhibit distinct temporal patterns, they share the same sets of model parameters, e.g.,

the same weight matrices in RNNs, the same convolution kernels in TCNs, or the same projection matrices in attentions. Some studies propose variable-specific modeling for RNNs. [Pan *et al.*, 2019] generates variable-specific weight matrices using additional meta-information, e.g., categories of points of interest around the deployed sensor locations. However, such additional meta-information may not be always available. [Bai *et al.*, 2020] learns variable-specific weight matrices using a purely data-driven manner without additional information. We empirically compare with it in the experiments. [Cirstea *et al.*, 2022b; Cirstea *et al.*, 2021] consider variable-specific modeling for networks other than RNNs, but not focusing on long term forecasting.

3 Preliminaries

Problem Definition. A multivariate time series records the values of N variables over time. $\mathbf{x}_t \in \mathbb{R}^N$ denotes the values of all N variables at timestamp t and $\mathbf{x}_t^{(i)} \in \mathbb{R}$ is the value of the i-th variable at t. Time series forecasting learns a function $\mathcal{F}$ that takes as into the observations in the historical H timestamps and predicts the future F timestamps.

$$\mathcal{F}_\phi(\mathbf{x}_{t-H+1}, ..., \mathbf{x}_{t-1}, \mathbf{x}_t) = (\hat{\mathbf{x}}_{t+1}, \hat{\mathbf{x}}_{t+2}, ..., \hat{\mathbf{x}}_{t+F}), \quad (1)$$

where ϕ denotes the learnable parameters of the forecasting model and $\hat{\mathbf{x}}_j$ is the predicted values at timestamp j.

Self-attention. Self-attention is a core operation in attention based models. Given a sequence, e.g., a time series, each timestamp attends to all timestamps in the time series, thus deriving a representation of the entire time series by capturing both long- and short-term dependencies. Consider a time series with H timestamps, e.g., time series $\mathbf{x}^{(i)} \in \mathbb{R}^H$ from the i-th variable. Canonical self-attention [Vaswani *et al.*, 2017] first transforms the time series into a query matrix $\mathbf{Q}^{(i)} = \mathbf{x}^{(i)}\mathbf{W}_Q \in \mathbb{R}^{H\times d}$, a key matrix $\mathbf{K}^{(i)} = \mathbf{x}^{(i)}\mathbf{W}_K \in \mathbb{R}^{H\times d}$ and a value matrix $\mathbf{V}^{(i)} = \mathbf{x}^{(i)}\mathbf{W}_V \in \mathbb{R}^{H\times d}$, where d represents the hidden representation and $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{d\times d}$ are *projection matrices*, which are learnable. Next, the output is represented as a weighted sum of the values in the value matrix $\mathbf{V}^{(i)}$, where the weights, a.k.a., attention scores, are computed based on the query matrix $\mathbf{Q}^{(i)}$ and the key matrix $\mathbf{K}^{(i)}$, as shown in Equation 2.

$$\mathcal{A}(\mathbf{Q}^{(i)}, \mathbf{K}^{(i)}, \mathbf{V}^{(i)}) = \varphi(\frac{\mathbf{Q}^{(i)}\mathbf{K}^{(i)T}}{\sqrt{d}})\mathbf{V}^{(i)}, \quad (2)$$

where φ represents the *softmax* activation function. Computing the attention scores requires quadratic $\mathcal{O}(H^2)$ time complexity and memory usage, which is a major drawback. Sparse versions of self-attentions exist, [Li *et al.*, 2019; Kitaev *et al.*, 2020; Zhou *et al.*, 2021], reducing the complexity to $\mathcal{O}(Hlog(H)^2)$ and $\mathcal{O}(H \cdot \log H)$. We strive for linear complexity. In addition, in related attention based works, the same projection matrices $\mathbf{W}_Q$, $\mathbf{W}_K$ and $\mathbf{W}_V$ are applied to all variables $\mathbf{x}^{(i)}$ for $1 \le i \le N$. In contrast, we strive for variable-specific projection matrices.

4 Triformer

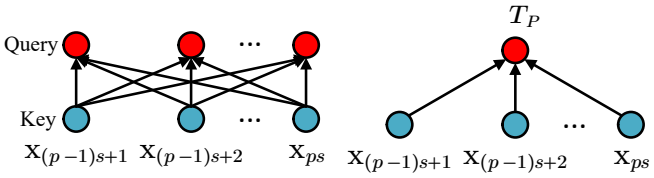
We propose Triformer for learning long-term and multi-scale dependencies in multivariate time series. An overview of the Triformer is shown in Figure 2. The design choices of Triformer are three-fold. First, we propose an efficient *Patch Attention* with linear complexity as the basic building block. Second, we propose a triangular structure when stacking multiple layers of patch attentions, such that the layer sizes shrink exponentially. This ensures linear complexity for multi-layer patch attentions and also enables extracting multi-scale features. Third, we propose a light-weight method to enable variable specific modeling, thus being able to capture distinct temporal patterns from different variables, without compromising efficiency.

4.1 Linear Patch Attention

To contend with the high complexity, we propose an efficient *Patch Attention* with linear complexity, to ensure competitive overall efficiency. Inspired by [Pan *et al.*, 2021], we break down the input time series of length H into $P = H/S$ patches along the temporal dimension, where S is the patch size. Figure 2 shows an example input time series of length $H = 12$ being split into $P = 4$ patches with patch size $S = 3$. We use $\mathbf{x}_p = \langle \mathbf{x}_{(p-1)\cdot S+1}, \ldots, \mathbf{x}_{p\cdot S} \rangle$ to denote the p-th patch.

Based on the patches, we compute attention scores per patch. If we naively use self-attention in a patch, we still have quadratic complexity (cf. Figure 3a).

To reduce the complexity to linear, for each patch p, we introduce a learnable, pseudo timestamp $\mathbf{T}_p \in \mathbb{R}^{N\times d}$ (cf. Figure 3b). The pseudo timestamp acts as a data placeholder where all the timestamps from the patch can write useful information which is then passed to the next layers. In Triformer, we choose to use the attention mechanism to update the pseudo timestamp, where the pseudo timestamp works as the Query in self-attentions. The pseudo timestamp queries all the "real" timestamps in the patch, thus only computing a single attention score for each real timestamp, giving rise to linear complexity. We call this Patch Attention (*PA*).



(a) Self-attention, $\mathcal{O}(S^2)$ (b) Patch attention, $\mathcal{O}(S)$

Figure 3: Computing self-attention (SA) vs. patch attention (*PA*) on patch p. In SA, each timestamp attends to all other timestamps, thus having $\mathcal{O}(S^2)$. In *PA*, each timestamp only attends to the pseudo timestamp, thus having $\mathcal{O}(S)$.

Equation 3 elaborates the computations of *PA* for pseudo timestamp $\mathbf{T}_p$. Note that $\mathbf{T}_p = \{\mathbf{T}_p^{(i)}\}_{i=1}^N$ and each variable has specific pseudo timestamp $\mathbf{T}_p^{(i)}$, thus being variable-specific.

$$\mathcal{PA}(\mathbf{T}_p, \mathbf{x}_p) = \left\{ \varphi(\frac{\mathbf{T}_p^{(i)}(\mathbf{x}_p^{(i)}\mathbf{W}_K)^T}{\sqrt{d}})(\mathbf{x}_p^{(i)}\mathbf{W}_V) \right\}_{i=1}^N \quad (3)$$

The reduced complexity of *PA* comes with a price that the temporal receptive filed of each timestamp is reduced to the

patch size S. In contrast, in the canonical attention, it is H covering all timestamps. This makes it harder to capture relationships among different patches and also the long-term dependencies, thus adversely affecting accuracy. To compensate for the reduced temporal receptive field, we introduce a recurrent connection (cf. the orange arrows in Figure 2) to connect the outputs of the patches, i.e., the updated pseudo timestamps according to Equation 3, such that the temporal information flow is maintained.

As gating mechanism is a crucial component for recurrent networks and has been showed to be powerful controlling the information flow [Dauphin *et al.*, 2017], we propose a gating rule in the recurrent connections in Equation 4.

$$\mathbf{T}_{p+1} = g(\mathbf{\Theta}_1\mathbf{T}_p + \mathbf{b}_1) \odot \sigma(\mathbf{\Theta}_2\mathbf{T}_p + \mathbf{b}_2) + \mathbf{T}_{p+1}, \quad (4)$$

where $\mathbf{\Theta}_1$, $\mathbf{\Theta}_2$, $\mathbf{b}_1$ and $\mathbf{b}_2$ are learned parameters for the recurrent gates, $\odot$ is element-wise product, $g(\cdot)$ is *tanh* activation function and $\sigma(\cdot)$ is a *sigmoid* function controlling the information ratio that is passed to the next pseudo timestamp.

4.2 Triangular Stacking

Stacking multiple layers of attention often help improve accuracy. In attention based models, each attention layer has the same input size, e.g., the input time series size H. In traditional self-attention same input and output have the same shape. Differently pooling-based methods are applying 1D convolution to the output to shrink the temporal horizon. When using `PA`s, we only feed the pseudo timestamps from the patches to the next layer, which shrinks the layer sizes exponentially. More specifically, the size of $(l+1)$-th layer is only $\frac{1}{S_l}$ of the size of the l-th layer, where S_l is the patch size of the l-th layer. This leads to Lemma 1.

Lemma 1. *An L-layer* `Triformer` *has a linear time complexity $\mathcal{O}(H)$ if patch size $S_l \geq 2$ where $1 \leq l \leq L$.*

Proof. Presented in the technical report [Cirstea *et al.*, 2022a].

In a multi-layer `Triformer`, each layer consists of different numbers of patches and thus having different number of outputs, i.e., pseudo timestamps. Instead of only using the last pseudo timestamp per layer, we aggregate all pseudo timestamps per layer into an aggregated output. More specifically, the aggregate output $\mathbf{O}^l$ at the l-th layer is defined as

$$\mathbf{O}^l = \theta^l(\mathbf{T}_1^l, .., \mathbf{T}_k^l, ..., \mathbf{T}_P^l), \quad (5)$$

where θ^l is a neural network, $\mathbf{T}_p^l$ denotes the pseudo timestamp for patch $p \in [1, P]$ at the l-th layer.

Finally, the aggregate outputs from all layers are connected to the predictor. This brings two benefits than just using the aggregate output of the last layer. First, the aggregate outputs represent features from different temporal scales, contributing to different temporal views. Second, it provides multiple gradient feedback short-paths, thus easing the learning processes. In the ablation study, we empirically justify this design choice (cf. Table 3 in Experiments).

Predictor. We use a fully connected neural network as the predictor due to its high efficiency w.r.t. longer term forecasting.

4.3 Variable-Specific Modeling

Variable-specific modeling can be achieved in a naive way by introducing different projection matrices for each variable, which leads to a very large parameter space. Figure 4 (a) shows that the naive approach needs to learn $N{\cdot}d^2$ parameters for each projection matrix. This may lead to over-fitting, incurs high memory usage, and does not scale well w.r.t. the number of variables N.

To contend with the above challenges, we propose a light-weight method to generate variable specific parameters. In addition, the method is purely data-driven that only relies on the time series themselves and does not require any additional prior knowledge. The overall process is illustrated in Figure 4 (b). First, we introduce a m-dimensional memory vector $\mathbf{M}^{(i)} \in \mathbb{R}^m$ for each variable with $i \in [1, N]$. The memory is randomly initialized and learnable. This makes the method purely data-driven and can learn the most prominent characteristics of each variable.

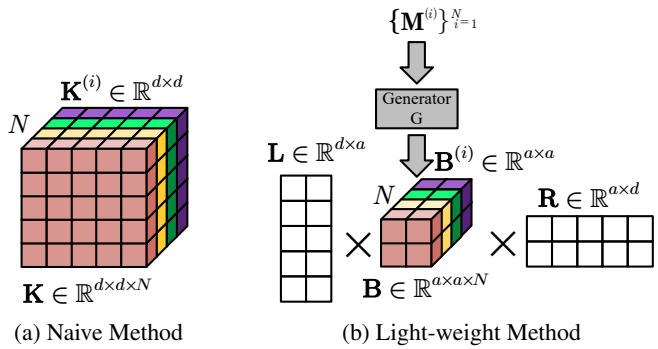


Figure 4: Enabling Variable-Specific Projection Matrices. Different colors represent matrices for different variables and $\mathbf{L}$ and $\mathbf{R}$ are variable-agnostic, thus without colors. For the i-th variable, the naive method learns directly matrix $\mathbf{K}^{(i)}$, while the light-weight method learns $\mathbf{K}^{(i)}$ by $\mathbf{L} \times \mathbf{B}^{(i)} \times \mathbf{R}$, where $\mathbf{B}^{(i)}$ is generated by memory $\mathbf{M}^{(i)}$ through a generator $\mathcal{G}$.

Next, we propose to factorize a projection matrix, e.g., a Key matrix $\mathbf{W}_K^{(i)} \in \mathbb{R}^{d\times d}$ into three matrices—a left variable-agnostic matrix $\mathbf{L}_K \in \mathbb{R}^{d\times a}$, a middle variable-specific matrix $\mathbf{B}^{(i)} \in \mathbb{R}^{a\times a}$, and a right variable-agnostic matrix $\mathbf{R}_K \in \mathbb{R}^{a\times d}$. We intend to make the middle matrix compact, i.e., $a \ll d$, thus making the method light-weight.

The left and right matrices are variable-agnostic, thus being shared with all variables. Different variables have their own middle matrices $\{\mathbf{B}^{(i)}\}_{i=1}^N$, thus making the middle matrices variable-specific. More specifically, the i-th variable's middle matrix $\mathbf{B}^{(i)}$ is generated from its memory $\mathbf{M}^{(i)}$ using a generator $\mathcal{G}(\cdot)$, e.g., a 1-layer neural network. This step is also essential to reduce the number of parameters to be learned. Learning a full matrix $\mathbf{B}$=$\{\mathbf{B}^{(i)}\}_{i=1}^N$ directly requires $N{\cdot}a^2$ parameters. In contrast, when using a generator, it requires $N{\cdot}m$ for the memories, and an additional overhead of $m{\cdot}a^2$ for the generator.

In addition to the reduced parameter sizes, the factorization also contributes to improved accuracy (cf. Table 3

Method		Reformer		LogTrans		StemGNN		AGCRN		Informer		Autoformer		Triformer	
	F	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE	MSE	MAE
$\mathbf{ETT}_{h_1}$	24	0.991	0.754	0.686	0.604	0.488	0.508	0.438	0.461	0.577	0.549	0.399	0.429	**0.328**	**0.380**
	48	1.313	0.906	0.766	0.757	0.473	0.505	0.488	0.486	0.685	0.625	0.405	0.439	**0.359**	**0.401**
	168	1.824	1.138	1.002	0.846	0.630	0.577	0.553	0.540	0.931	0.752	0.498	0.486	**0.433**	**0.449**
	336	2.117	1.280	1.362	0.952	0.658	0.598	0.611	0.566	1.128	0.873	0.500	0.480	**0.487**	**0.475**
	720	2.415	1.520	1.397	1.291	0.764	0.675	0.778	0.643	1.215	0.896	0.498	0.500	**0.488**	**0.475**
$\mathbf{ETT}_{m_1}$	24	0.724	0.607	0.419	0.412	0.302	0.381	0.318	0.358	0.323	0.369	0.377	0.417	**0.252**	**0.329**
	48	1.098	0.777	0.507	0.583	0.382	0.436	0.426	0.429	0.494	0.503	0.429	0.442	**0.275**	**0.337**
	96	1.433	0.945	0.768	0.792	0.419	0.461	0.409	0.433	0.678	0.614	0.458	0.460	**0.314**	**0.371**
	288	1.820	1.094	1.462	1.320	0.522	0.518	0.545	0.531	1.056	0.786	0.632	0.526	**0.385**	**0.427**
	672	2.187	1.232	1.669	1.461	0.644	0.590	0.567	0.570	1.192	0.926	0.602	0.540	**0.437**	**0.448**
Weather	24	0.655	0.583	0.435	0.477	0.377	0.416	0.947	0.774	0.335	0.381	0.408	0.445	**0.323**	**0.364**
	48	0.729	0.666	0.426	0.495	0.438	0.467	0.948	0.777	0.395	0.459	0.475	0.487	**0.390**	**0.429**
	168	1.318	0.855	0.727	0.671	0.554	0.545	0.950	0.774	0.608	0.567	0.576	0.550	**0.497**	**0.501**
	336	1.930	1.167	0.754	0.670	0.598	0.573	0.946	0.772	0.702	0.620	0.593	0.558	**0.538**	**0.531**
	720	2.726	1.575	0.885	0.773	0.676	0.619	0.938	0.770	0.831	0.731	0.682	0.634	**0.587**	**0.563**
ECL	48	1.404	0.999	0.355	0.418	0.209	0.311	0.385	0.398	0.344	0.393	0.193	0.310	**0.183**	**0.279**
	168	1.515	1.069	0.368	0.432	0.265	0.355	0.481	0.509	0.368	0.424	0.223	0.334	**0.182**	**0.288**
	336	1.601	1.104	0.373	0.439	0.291	0.380	0.564	0.561	0.381	0.431	0.232	0.345	**0.202**	**0.309**
	720	2.009	1.170	0.409	0.454	0.317	0.400	0.725	0.678	0.406	0.443	0.257	0.361	**0.251**	**0.335**
	960	2.141	1.387	0.477	0.589	0.329	0.411	1.005	0.829	0.460	0.548	0.265	0.364	**0.248**	**0.339**

Table 2: Overall accuracy. **Bold** highlights the best results. Underline highlights the second best results.

in Experiments). Since all variables' time series share the variable-agnostic matrices $\mathbf{L}$ and $\mathbf{R}$, they act as an implicit regularizer—the amount of combinations that can be produced using them is limited. It also facilitates implicit knowledge sharing among variables.

Equation 6 summarizes the generation of variable-specific Key and Value matrices $\mathbf{W}_K^{(i)}$ and $\mathbf{W}_V^{(i)}$, which replace $\mathbf{W}_K$ and $\mathbf{W}_V$ in Equation 3 to make Patch Attention variable-specific. We do not need the Query matrix as *PA* does not need it and employs the pseudo timestamp instead.

$$\begin{bmatrix} \mathbf{W}_K^{(i)} \\ \mathbf{W}_V^{(i)} \end{bmatrix} = \begin{bmatrix} \mathbf{L}_K \mathcal{G}(\mathbf{M}^{(i)}) \mathbf{R}_K \\ \mathbf{L}_V \mathcal{G}(\mathbf{M}^{(i)}) \mathbf{R}_V \end{bmatrix} \tag{6}$$

5 Experiments

We report on a comprehensive empirical study on four real-world, commonly used time series long term forecasting data sets to justify our design choices and demonstrate that `Triformer` outperforms the state-of-the-art methods.

5.1 Experimental Setup

Datasets: We consider four data sets and follow the setup from the-state-of-the-art method for long term forecasting [Zhou *et al.*, 2021].

ETT $_{h_1}$, **ETT**$_{m_1}$ (Electricity Transformer Temperature)[1]: **ETT**$_{h_1}$ and **ETT**$_{m_1}$ has observations every 15 minutes. Each observation consists of 6 power load features, making them 6-variate time series. To ensure fair comparisons with existing studies, following [Zhou *et al.*, 2021], the train/validation/test data cover 12/4/4 months.

[1]https://github.com/zhouhaoyi/ETDataset

ECL (Electricity Consumption Load)[2] is a 321-variate time series, which records the hourly electricity consumption (Kwh) of 321 clients. Following [Zhou *et al.*, 2021], the train/validation/test data cover 15/3/4 months.

Weather [3] is a 12-variate time series which record 12 different climate features, e.g., temperature, humidity, etc, from 1,600 U.S. locations. The data is collected hourly. Following [Zhou *et al.*, 2021], the train/validation/test data cover 28/10/10 months.

Forecasting Setups: We use historical H timestamps to forecast the future F timestamps. We vary H and F for different data sets, by following a commonly used long term forecasting setup [Zhou *et al.*, 2021]. We vary F progressively in {24, 48, 168, 336, 720, 960} for the hourly data sets **ETT**$_{h_1}$, **ECL**, and **Weather**, corresponding to 1, 2, 7, 14, 30, and 40 days ahead; and we vary F in {24, 48, 96, 288, 672} for **ETT**$_{m_1}$ corresponding to 6, 12, 24, 72 and 168 hours ahead. We also vary the input historical timestamps H. For **ETT**$_{h_1}$, **Weather** and **ECL**, we vary H from {24, 48, 96, 168, 336, 720}, and from {24, 48, 96, 192, 288, 672} for **ETT**$_{m_1}$. Following [Zhou *et al.*, 2021], for each F value, we iterate all possible H values and report the best values in Table 2. A more detailed description of the setup can be found in the technical report [Cirstea *et al.*, 2022a].

Baselines: We select six recent and strong baselines from different categories shown in Table 1, including `StemGNN` [Cao *et al.*, 2020], `AGCRN` [Bai *et al.*, 2020], `Informer` [Zhou *et al.*, 2021], `Reformer` [Kitaev *et al.*, 2020], `LogTrans` [Li *et al.*, 2019], and `Autoformer` [Wu *et al.*, 2021a].

[2]https://archive.ics.uci.edu/ml/datasets/ElectricityLoadDiagrams20112014

[3]https://ncei.noaa.gov/data/local-climatological-data

Implementation Details: Due to space limitation the implementation details are listed in the technical report [Cirstea *et al.*, 2022a].

5.2 Experimental Results

Table 2 shows the overall accuracy. The results of `Informer`, `Reformer` and `LogTrans` are collected from [Zhou *et al.*, 2021]. For `AGCRN`, `StemGNN`, and `Autoformer` we have used their original implementations which are publicly available. `Triformer` outperforms all baselines on all datasets. In most settings, `AGCRN` achieves better results when compared to the three attention based methods that are variable-agnostic `Reformer`, `LogTrans` and `Informer`. This highlights the needs of variable-specific modeling. `Autoformer` achieves the second best accuracy which highlights the importance of auto-correlation. Finally our proposed method, `Triformer`, achieves the best accuracy in all settings.

Ablation Study: We perform an ablation study on the **ECL** dataset, as shown in Table 3. First, we remove the variable-specific modeling (VSM), where the same projection matrices are shared across all variables. We observe a significant loss of accuracy, which well justifies our design considerations to capture distinct temporal patterns of different variables. A shared parameter space for all variables fails to do so. Second, we replace the light-weight variable specific modeling by the naive way (cf. Figure 4(a)). We observe that the naive method is less accurate and incurs significant more parameters to be learned, which are well aligned with our design considerations. The generator $\mathcal{G}$ in `Triformer` takes very limited additional time to generate matrix $\mathbf{B}$. Third, we do not stack multiple layers patch attentions but use only 1 layer of patch attentions. We observe a significant drop in accuracy, suggesting that the proposed multi-layer, triangular structure, is very effective. Fourth, we remove the multi-scale modeling such that only the top most layer's output, rather than the outputs from all layers, is fed into the predictor. The accuracy drops significantly, which justifies our design choices of using multi-scale representations in the predictor. Finally, we remove the recurrent connections that connect two consecutive pseudo timestamps within *PA*s. The temporal information flow is thus broken, meaning that each patch is computed independently of the others. This results in sub-optimal accuracy. However, we note that the accuracy loss is much smaller than removing other components. One of the benefit of this variant is that this makes the computations of all *PA*s per layer fully parallelizable thus improving the running time.

	MSE	MAE	#Param	s/epoch
`Triformer`	0.183	0.279	347k	73.11
w/o VSM	0.215	0.273	303k	33.25
w Naive VSM	0.191	0.288	826k	63.81
w/o Stacking	0.203	0.295	258k	56.26
w/o Multiscale	0.266	0.352	285k	71.58
w/o Recurrent	0.191	0.290	346k	65.75

Table 3: Ablation Study, **ECL**.

Impact of m: To study the impact of m, i.e., the memory size of $\mathbf{M}^{(i)}$ (cf. the section "Variable-Specific Modeling" and Figure 5 in the main paper), we ran multiple experiments in which we vary m among $\{3, 5, 16, 32\}$. The results are shown in Table 4. We observe insignificant variations in terms of accuracy, which indicates the proposed method is insensitive w.r.t. the memory size.

m	MSE	MAE
3	0.185	0.281
5	**0.183**	**0.279**
16	0.186	0.285
32	0.188	0.284

Table 4: Effect of m, **ECL**.

Impact of a: We study the impact of a, i.e., the size of the middle matrix $\mathbf{B}^{(i)}$ (cf. the section "Variable-Specific Modeling" and Figure 4b). We vary a among the following values $\{3, 5, 16, 32\}$ and the results are shown in Table 5. We observe that when a is too small, e.g., $a = 3$, the accuracy drops significantly, which is understandable as the variable-specific matrix $\mathbf{B}^{(i)}$ has a shape of $a \times a$, which may be too compact to capture sufficient variable-specific temporal patterns. In addition, we observe that $a = 5$ is sufficient and increasing further a to 32 does not contribute to improved accuracy.

a	MSE	MAE
3	0.191	0.288
5	**0.183**	**0.279**
16	0.184	0.280
32	0.186	0.281

Table 5: Effect of a, **ECL**.

Efficiency: Figure 5 shows the training runtime (seconds/epoch) of `Triformer` against other four baselines that show the second best accuracy in some data sets, i.e., `Informer`, `AGCRN`, `StemGNN` and `Autoformer`, as shown in Table 2. When varying H: (i) `Triformer` is faster than `Informer` and `Autoformer`, which is in accordance with the complexities of the two methods and also due to the efficient fully connected neural network based predictor. (ii) `StemGNN` increases faster as H increases, while `Triformer` almost keeps steady. (iii) we observe that the recurrent based method `AGCRN` falls behind when compared with attention based methods, especially when the input time series is long, i.e., large H. When varying F, `Triformer` is the fastest in all settings. In addition, for inference time, all methods are less than 13.3 ms per inference, which are sufficiently efficient to support real-time forecasting.

Visualization of Learned Memories: We visualize the learned memories $\mathbf{M}^{(i)}$ to investigate whether they may capture distinct and the most prominent patterns of different variable's time series. We select 8 time series from the **ECL** data sets, as shown in Figure 6. We use `t-SNE` [Maaten and Hinton, 2008] to compress each variable memory $\mathbf{M}^{(i)}$ to a 2D point, which is also shown in Figure 6. We observe that the

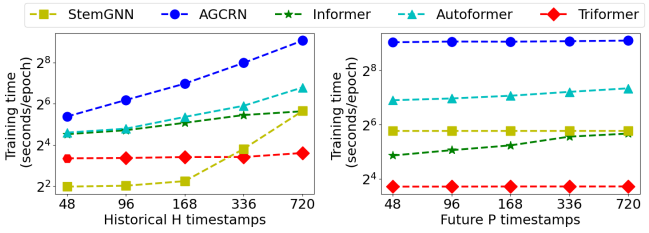


Figure 5: Training Runtime, seconds/epoch

points are spread over the space, indicating that different time series have their own unique patterns. In addition, we observe that when the variables have similar temporal patterns, their learned memories are close by. For example, the time series of variables 6, 7, and 8 are similar, and their corresponding memories are also clustered together, i.e., in the right, top corner.

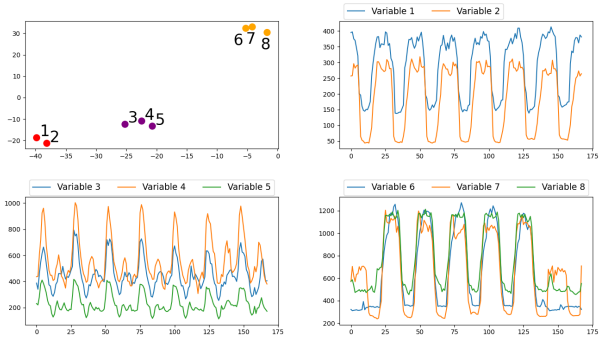


Figure 6: Visualization of the Learned Memories $\mathbf{M}^{(i)}$ for different variables along with their time series, **ECL**

Additional experiments for very long sequence forecasting, hyper-parameter sensitivity analysis for number of layers and different patches sizes, and empirical evidence that the proposed `VSM` can generalize to existing methods are presented in the technical report [Cirstea *et al.*, 2022a].

6 Conclusion and Outlook

We propose `Triformer`, a triangular structure that employs novel patch attentions, which ensures linear complexity. Furthermore, we propose a light-weight method to generate variable-specific projection matrices which are tailored to capture distinct temporal patterns for each variable's time series. Extensive experiments on four data sets show that our proposal outperforms other state-of-the-art methods for long sequence multivariate time series forecasting. In future work, it is of interest to explore different ways of supporting dynamic input lengths and to enhance model training using curriculum learning [Yang *et al.*, 2021; Yang *et al.*, 2022].

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